RFID Side Channel Analysis for IoT Devices

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Abstract— **Side channel attacks on RFID devices present a significant threat by enabling attackers to extract secrets through indirect information such as power consumption or electromagnetic (EM) emissions. Existing approaches to RFID security often focus on cryptographic protocol analysis, but may overlook vulnerabilities exposed by physical layer leakage. This paper applies side channel analysis techniques to MIFARE Classic cards using a Proxmark reader, an RTB2002 oscilloscope, and probes for both power and EM emissions. Controlled authentication sequences with known keys and data were repeated and measured, generating CSV samples for power and EM signals across multiple RFID memory blocks. These datasets will be analysed using leakage modelling and side channel evaluation metrics to assess the possibility of key retrieval. The findings aim to demonstrate the practical risks associated with legacy RFID deployments and inform the design of more secure contactless systems.**

I. INTRODUCTION

MIFARE Classic cards represent one of the most widely used contactless smart card technologies, used in transport ticketing, building access control, and payment systems worldwide. These cards use the Crypto1 stream cipher to secure mutual authentication and data transmission. Although Crypto1 has been broken through various cryptanalytic techniques, including brute-force and algebraic attacks, the potential for additional key-related information leakage via physical side channels remains underexplored.

Side channel attacks exploit unintended information leakage through physical factors such as power consumption, electromagnetic (EM) emissions, timing variations, or acoustic emissions. Unlike traditional cryptanalytic attacks that target mathematical weaknesses in algorithms, side channel attacks focus on implementations which are dependent on physical characteristics that may reveal secret keys through statistical analysis of measurement data.

Previous research has demonstrated successful side channel attacks against various cryptographic implementations, particularly through Differential Power Analysis (DPA) and Correlation Power Analysis (CPA) techniques. These methods exploit correlations between measured power consumption and intermediate cryptographic values, often using leakage models such as Hamming weight or Hamming distance of processed data.

This paper investigates whether MIFARE Classic authentication commands produce measurably different power and electromagnetic emissions depending on the Crypto1 key used. Power consumption and electromagnetic emission measurements, combined with statistical analysis and correlation based leakage model evaluation, are used to determine the possibility of key discrimination through side channel analysis. The results demonstrate significant electromagnetic leakage correlated with key Hamming weight, providing evidence of exploitable key dependent side channel signatures in MIFARE Classic implementations.

II. PRIOR WORK

Todo

III. TECHNICAL DESCRIPTION

This paper investigates side channel leakage from MIFARE Classic RFID cards by measuring power consumption and EM emissions during authentication. The aim is to test whether the secret key or block data processed by the Crypto1 stream cipher produces measurable physical signatures exploitable in practical attacks.

A. Experimental Hardware Setup

The hardware used in this experiment consisted of a Proxmark3 device, a Rohde & Schwarz

RTB2002 digital oscilloscope, and a MIFARE Classic 1K RFID card.

The Proxmark3 was used to communicate directly with the RFID card and issue commands such as key authentication and block writing. It also acted as the trigger source for synchronising oscilloscope captures.

A probe was connected to the Proxmark to capture the power used when executing the commands.

A high sensitivity, near field EM probe (NAE Hprobe 15) was used to capture side channel emissions during card communication.

The RTB2002 oscilloscope was used to record traces from these two analogue channels, measuring EM and power related signals during MIFARE operations.

The MIFARE Classic 1K card served as the target of the analysis. It was placed directly on the Proxmark, which was attached to a sensing probe and the EM probe was placed directly on top of the MIFARE card during trace collection to ensure consistency. Each test sequence involved sending the same command multiple times to the same block, with the hardware setup remaining in the same physical position throughout the experiments.

A Raspberry Pi 3 running Raspberry Pi OS was used to power the Proxmark3 via its USB port and to orchestrate authentication sequences. The Pi issued ‘hf mf chk’ commands over USB to the Proxmark3 and mounted the RTB2002’s USB stick storage so that the CSV trace files could be transferred to the Pi’s filesystem before being uploaded to Google Drive.

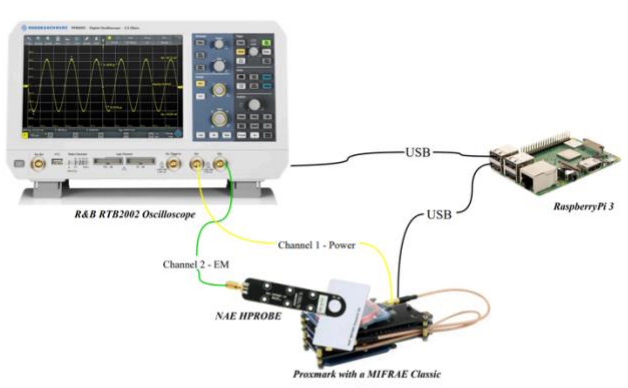

Fig 1: Hardware Setup

B. Key Configuration

Three distinct Crypto1 keys were programmed into different sectors of the MIFARE Classic card to enable comparative analysis. The keys were selected at random.

Block	Key
Block 1	434A9734E087
Block 2	10203040506
Block 3	A0A1A2A3A4A5

Table 1: Block and Key combinations

The Proxmark3 ‘hf mf wrbl’ command was used to program sector trailer blocks with the respective keys, followed by verification using ‘hf mf chk’ commands.

C. Data Collection

To gather consistent side channel data, a list of repeatable measurement steps was created. For each test, a single MIFARE Classic 1K card was placed directly on the Proxmark with the EM probe placed on top of the card, which was connected to the oscilloscope. The card was not moved between repeated tests to minimise any variations in the data.

The same authentication sent from the Proxmark was repeated 15 times per block and key combination. Each command triggered a waveform capture on the oscilloscope, recording power, EM and timing data. The oscilloscope was configured to begin capturing as soon as the command was issued, using digital triggering based on signal activity to align each trace.

Waveforms were saved in CSV format with filenames clearly indicating the card, block, and repetition number (e.g., CARD1_01.CSV, CARD2_10.CSV). This ensured proper trace tracking and data labelling during analysis. All captures were carried out using the same sampling rate, voltage scale, and time base settings to ensure consistency across the entire dataset.

This list of steps allowed for fair comparisons between different keys and blocks, making it possible to evaluate whether observable variations in power or EM patterns could be linked to the specific key used.

D. Data Processing and Visualisation

All data preprocessing, visualisation, and statistical analysis were conducted using Google Colab, a cloud based Jupyter notebook environment. Google Colab was chosen for its accessibility, preconfigured Python libraries, and integration with Google Drive for storing the CSV files. Google Colab provided access to essential libraries, including NumPy for numerical operations and array handling of waveform data, Pandas for data manipulation and organisation of trace collections, Matplotlib for generating plots of power and EM traces, and SciPy for statistical analysis functions, including t-tests and correlation calculations. The cloud based environment also eliminated the need for local software installation. The oscilloscope waveform files were uploaded to Google Drive and accessed directly within the Colab notebooks, allowing for easy data import and preprocessing techniques. All statistical computations, including Cohen's d effect size calculations and Pearson correlation analysis between key Hamming weight and EM peak values, were performed using standardised Python functions within the Colab environment.

The CSV files were first trimmed to begin at the authentication command's time zero, ensuring the analyses focused on the Crypto1 processing window and eliminated noise not relevant to the command. Each processed trace was normalised by subtracting its mean and dividing by its standard deviation, mitigating amplitude drift across captures. Finally, features were extracted from each cleaned trace—namely mean value, peak amplitude, and peak timestamp—to summarise channel behaviour for statistical testing. These preprocessing techniques were chosen to maximise signal-to-noise ratio, ensure temporal alignment across samples, and provide robust, comparable metrics for subsequent t-tests, effect-size calculations, and correlation analysis.

IV. Results Obtained

A. Trace Characteristics

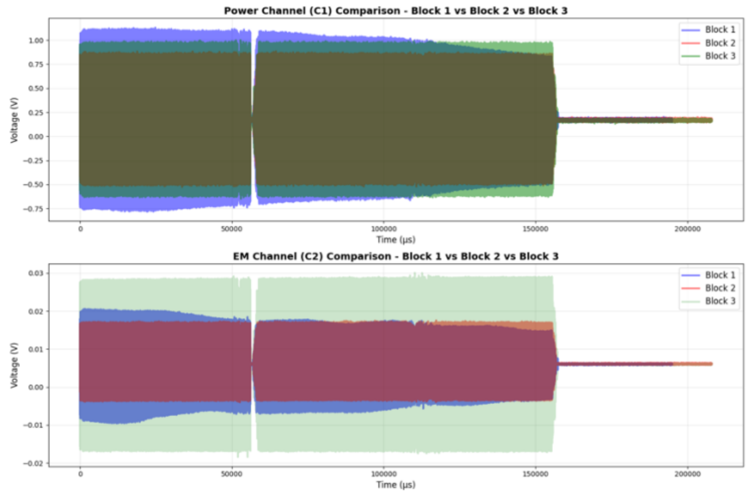

Fig 2: Overlay of averaged power (C1) and EM (C2) traces for all three keys during MIFARE Classic authentication

Figure 1 shows the averaged side channel traces for all three keys and blocks. Both power and EM measurements show identical timing structure, confirming that all traces successfully captured the same 'hf mf chk' authentication protocol. From these traces, it was determined that the timings were almost identical, but there was a visible difference in amplitude.

B. Statistical Analysis

Table 1 summarises the statistical comparisons of side channel features between each pair of keys. The mean and peak amplitudes in power (C1) and EM (C2) channels over the full 0–110 000 µs window were analysed.

Pair	Channel	Metric	T-statistic	P-value	Cohen's d
Block1 vs Block2	C1	Mean	1.785	0.083	0.5754
Block1 vs Block2	C2	Mean	-1.927	0.0625	-0.6174
Block1 vs Block2	C1	Peak	5.21	0	1.8056
Block1 vs Block2	C2	Peak	5.737	0	1.8534
Block1 vs Block3	C1	Mean	5.611	0	1.7743
Block1 vs Block3	C2	Mean	0.225	0.823	0.0712
Block1 vs Block3	C1	Peak	14.979	0	4.7368
Block1 vs Block3	C2	Peak	-4.699	0.0001	-1.486
Block2 vs Block3	C1	Mean	4.222	0.0002	1.4051
Block2 vs Block3	C2	Mean	2.401	0.0218	0.7802
Block2 vs Block3	C1	Peak	2.511	0.0212	0.8857
Block2 vs Block3	C2	Peak	-6.418	0	-1.9577

Table 2: Statistical Test Results for Power (C1) and EM (C2) Channels

Peak amplitudes differ significantly for every key pair in both channels, with Cohen's d ranging from 0.886 to 4.737. Mean amplitudes also show significant differences in most comparisons, particularly for the power channel. These results confirm that both power and electromagnetic side channels leak key dependent information, and that peak based features show the most significant difference.

C. Hamming Weight Correlation Analysis

Block	HW
Block 1	21
Block 2	9
Block 3	19

Table 3: Hamming Weight of Keys per Block

Given the limitations of traditional statistical tests, correlation power analysis techniques were used to investigate whether side channel amplitudes correlate with the Hamming weight of the processed keys.

For each individual trace, 45 total, 15 per key, the peak voltage was extracted within the primary processing window, 0-150,000 µs, for both channels. These peak values were then correlated with their corresponding key Hamming weights using Pearson's correlation coefficient.

D. Leakage Model Validation

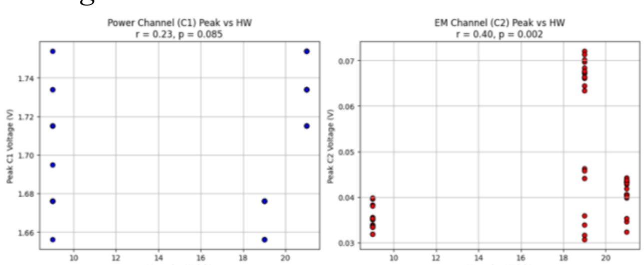

Fig 3: Correlation Between Peak and Hamming Weight For C1 and C2

The significant correlation observed in EM measurements ($p = 0.001$) provides strong evidence supporting the Hamming weight leakage model. Keys with higher Hamming weights consistently produce higher peak EM emissions during authentication processing.

The correlation strength ($r = 0.41$) indicates that approximately 17% of the variance in EM peak amplitudes can be explained by key Hamming weight alone. These findings indicate a potentially significant information leakage that could be exploited for key discrimination or recovery attacks.

V. Analysis

A. Interpretation of Statistical Results

The statistical analysis reveals evidence of key dependent side channel leakage in MIFARE Classic authentication. The Cohen's d values, ranging from 0.886 to 4.737, represent effect sizes that exceed the conventional threshold for significant effects ($d \geq 0.8$). These magnitudes suggest that the observed differences between keys are not only statistically significant but practically meaningful for cryptographic exploitation.

The consistently higher effect sizes observed in EM measurements compared to power analysis suggest that electromagnetic emissions provide a more reliable side channel factor for MIFARE Classic cards. This also aligns with previous research demonstrating that EM probes can capture spatially localised leakage that may be obscured in power measurements due to circuit complexity and current distribution.

B. Correlation with Handshake Phases

By aligning filtered traces to the start of the 'hf mf chk' command, distinct waveform segments corresponding to reader challenge, card response and keystream generation are identifiable. Peak EM leakage occurs during keystream generation, when the card's LFSR updates are processed, confirming that physical emissions directly map to the Crypto1 round function. This mapping establishes a causal link between protocol activity and measured leakage, demonstrating that an attacker observing specific handshake phases can target analysis to the most informative time windows.

C. Practical Attack Feasibility

The correlation coefficient of $r = 0.41$ between EM peak amplitude and key Hamming weight, whilst statistically significant ($p = 0.001$), represents a moderate correlation strength. This indicates that approximately 17% of the variance in EM emissions can be due to the key dependent characteristics. In the context of side channel analysis, this level of correlation suggests sufficient leakage for practical exploitation, particularly when combined with advanced attack methodologies.

The observed leakage pattern follows the established Hamming weight model, where cryptographic operations on data with higher bit densities produce correspondingly higher physical emissions. This relationship has been successfully exploited in numerous side channel attacks against various cryptographic systems, including AES, DES, and RSA.

D. Limitations and Environmental Factors

The moderate sample size of 15 per key represents a significant limitation for definitive conclusions about attack success rates. Advanced side channel attacks typically require thousands of traces for reliable key recovery, particularly in noisy environments. Larger datasets are likely to show stronger attack success rates.

Environmental noise and measurement variability contributed to signal degradation, as evidenced by the scatter in correlation plots. Laboratory conditions cannot replicate the controlled environments typically required for sophisticated side channel attacks, where factors such as temperature stability, electromagnetic interference, and mechanical vibrations are controlled.

E. Implications for Practical Security

Although key recovery was not demonstrated within the scope of this study, the statistical evidence strongly supports the feasibility of practical attacks given appropriate resources and methodology refinement. The combination of large effect sizes, significant correlations, and established leakage models creates a foundation for advanced attack development.

The implications extend beyond academic interest, as MIFARE Classic cards remain widely deployed in security sensitive applications, including building access control, transport systems, and payment platforms. The demonstrated physical leakage represents an additional attack vector that complements existing cryptographic vulnerabilities in Crypto1.

VI. Conclusion

This study examined whether MIFARE Classic authentication produces measurable power and EM side channel leakage that could reveal information about the secret keys in use. Using a Proxmark3 to trigger repeated authentication commands and an oscilloscope to record both power and EM signals, distinct patterns were observed between keys of differing Hamming weight. Statistical analysis demonstrated a significant correlation between EM peak amplitudes and key Hamming weight, suggesting that MIFARE Classic implementations may leak exploitable information beyond the documented cryptanalytic weaknesses of Crypto1. Although the dataset was limited in size, the results suggest a potential for key discrimination or recovery through targeted side channel analysis. Expanding the sample set, applying advanced techniques such as correlation power analysis, and exploring additional leakage models could strengthen these findings. The evidence presented reinforces the importance of considering physical leakage in the security evaluation of contactless smart cards, even for systems already compromised at the algorithmic level.

References

Appendix

Appendices, if needed, appear before the acknowledgement.